

Nicholas Karantakis

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Education

University of Toronto | Cumulative GPA 3.91/4.0

Sept. 2023 – Apr. 2028

Bachelor of Applied Science and Engineering (B.A.Sc.) in Mechanical Engineering + PEY Co-op

Toronto ON

Intended Minors: Robotics and Engineering Business

Experience

Hardware Vision Engineering Intern

Sept. 2025 – Present

Martinrea Advanced Manufacturing Technology Group

Vaughan ON

- Designed and installed **computer vision system** for part detection into **automotive manufacturing** lines.
- Aided along every step of process including building vision panels, data collection, model training, hardware installation, lighting configuration UI development, model testing, configuring network/IP settings and more.
- Working on **R&D projects** within the team to enhance computer vision systems for various applications in manufacturing.

Lead-Free Piezoelectric Nanogenerator Research

May. 2025 – Aug. 2025

Mechatronics and Microsystems Design Laboratory

Toronto ON

- Conducted research on lead-free piezoelectric nanogenerators (PENG) by **doping** and **synthesizing** Zinc Oxide nanowires with various elements to enhance piezoelectric efficiency for **sustainable energy harvesting**.
- Designed and fabricated custom testing apparatus using **SolidWorks** and in-house **machining** tools; performed **soldering** and assembly to support voltage output measurements of doped Zinc Oxide nanowires.
- Developed and optimized experimental setups for **electrical characterization** of piezoelectric materials, resolving technical issues and refining testing methodologies to **improve data accuracy** and **reproducibility**.

Airframe Member

Jan. 2024 – May. 2025

University of Toronto Aerospace Team UAS AEAC Drone Team

Toronto ON

- Facilitated the **design, building, and testing** of airframe components for the AEAC competition.
- Developed and tested **engineered point of failure** for motor mounts to protect propellers in case of a crash.
- Created design for drone shell in **SolidWorks** using **surface modeling** and developed mold for fiberglass layups

Lung Airway and Vasculature Research

May. 2024 – Aug. 2024

Latner Thoracic Research Laboratories and Bazylak Group

Toronto ON

- Optimized MATLAB vessel network extraction pipeline using MATLAB Profiler tools and memory allocation techniques to achieve **47% reduction in process time** for extracting the geometric characteristics of lung vasculature and airways from image stacks.
- Developed **imaging techniques** for mouse lung airways based on casting procedures from literature using MICROFIL and silicone rubber casting material to produce micro-computed tomography images with **over 1000 branch** segments up to **~30 generations**.
- Constructed **simplified airway geometries** for **Computational Fluid Dynamics analysis** from image data using Python and Blender.

Production Support

July. 2022 – Aug. 2022

Mitsubishi Heavy Industries Canadian Aerospace

Mississauga ON

- Contributed to **Bombardier Global Express/5000** wings and center fuselage production.
- Supported various tasks, including paint touchups, cleaning, sandblasting, and preparing wings for leak tests.
- Enhanced soft skills such as adaptability and communication by assuming diverse roles at all production stages including the **spar shop, final inspection**, and **preparation for shipping**.

Skills

Programming: MATLAB, Python, JavaScript, Arduino C++, HTML/CSS

Frameworks: React, Node.js, Flask

Libraries: Git, Pandas, NumPy, Matplotlib, Network X, Media Pipe, Open CV, PyQt5, Mapbox, YOLO,

CAD / 3D Modelling: SolidWorks CSWA, Blender, 3D Slicer

Machining: Lathe, Mill, Drill Press, Laser Cutting

Office Tools: Excel, Word, PowerPoint

Personal: Public Speaking, Adaptability, Problem-Solving, Teamwork